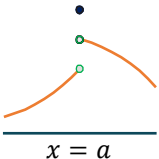




Monotonicity : RACE-01

Single Correct Question

1. $f(x) = |x - 1| + 2|x - 3| - |x + 2|$ is monotonically increasing at $x = ?$
 (A) -2 (B) 0 (C) 3 (D) 5 MMN125
2. For $0 \leq x \leq 1$, the function $f(x) = |x| + |x - 1|$ is-
 (A) Monotonically increasing (B) Monotonically decreasing
 (C) Constant function (D) Identity function MMN126
3. The function $y = \frac{2x^2 - 1}{x^4}$ is
 (A) Always increasing (B) Always decreasing
 (C) Neither increasing nor decreasing (D) None of these MMN127
4. In the following graph comment on monotonicity of $f(x)$ at $x = a$.

 (A) Monotonically increasing at $x = a$
 (B) Monotonically decreasing at $x = a$
 (C) neither monotonically increasing nor decreasing at $x = a$
 (D) Can't say at $x = a$ MMN128
5. When $x \in (0, 1)$, function $f(x) = 1/\sqrt{x}$ is -
 (A) Increasing (B) Decreasing
 (C) Neither increasing nor decreasing (D) Constant MMN129
6. Let f be the function $f(x) = \cos x - \left(1 - \frac{x^2}{2}\right)$ then $f(x)$ is strictly increasing in the interval
 (A) $(-\infty, \infty)$ (B) $(-2, \infty)$ (C) $[0, \infty)$ (D) $(0, \infty)$ MMN130
7. Function $f(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$ is-
 (A) Increasing (B) Decreasing
 (C) Neither increasing nor decreasing (D) Even function MMN131
8. Function $f(x) = \ln \sin x$ is monotonically increasing when-
 (A) $x \in (\pi/2, \pi)$ (B) $x \in (-\pi/2, 0)$ (C) $x \in (0, \pi)$ (D) $x \in (0, \pi/2)$ MMN132

9. For what value of 'a' the function $f(x) = x + \cos x - a$ increases
 (A) 0 (B) 1 (C) -1 (D) Any value **MMN133**
10. Let $f(x) = x^3 + ax^2 + bx + 5 \sin^2 x$ be an increasing function in the set of real numbers R . Then a & b satisfy the condition:
 (A) $a^2 - 3b - 15 > 0$ (B) $a^2 - 3b + 15 \leq 0$
 (C) $a^2 + 3b - 15 < 0$ (D) $a > 0$ & $b > 0$ **MMN134**

Multiple Correct Question

11. Let $f(x) = \begin{cases} x & x \in [1,2] \\ 5-x & x \in (2,4) \\ 2 & x = 4 \\ 7-x & x \in (4,6] \end{cases}$ then which of the following statement is/are correct about $f(x)$?
 (A) Function is strictly increasing at point $x = 2$
 (B) Function is strictly increasing at point $x = 4$
 (C) Function is not increasing at point $x = 2$ and $x = 4$
 (D) Function is not increasing at point $x = 1$ **MMN135**

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	C	C	C	B	C	A	D	D	B	A,B



Monotonicity : RACE-02

Single Correct Question

1. The greatest, the least values of the function, $f(x) = 2 - \sqrt{1 + 2x + x^2}$, $x \in [-2, 1]$ are respectively
 (A) 2, 1 (B) 2, -1 (C) 2, 0 (D) -2, 3

MMN136

2. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is

- (A) $[0, 1]$ (B) $\left(0, \frac{1}{2}\right]$ (C) $\left[\frac{1}{2}, 1\right]$ (D) $(0, 1]$

MMN137

3. Which of the following is true

- (A) $1 + x^2 > (x \sin x + \cos x)$ for $x \in [0, \infty)$ (B) $1 + x^2 \geq (x \sin x + \cos x)$ for $x \in [0, \infty)$
 (C) $1 + x^2 \leq (x \sin x + \cos x)$ for $x \in [0, \infty)$ (D) $1 + x^2 < (x \sin x + \cos x)$ for $x \in [0, \infty)$

MMN138

4. Which of the following is true

- (A) $\sin x - \sin 2x \leq 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$ (B) $\sin x - \sin 2x \geq 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$
 (C) $\sin x - \sin 2x \leq 2x$ for all $x \in \left[-\frac{\pi}{3}, 0\right]$ (D) $\sin x - \sin 2x > 2x$ for all $x \in \left[0, \frac{\pi}{3}\right]$

MMN139

5. Which of the following is true

- (A) $\frac{x^2}{2} + 2x + 3 > (3 - x)e^x$ for all $x \geq 0$ (B) $\frac{x^2}{2} + 2x + 3 < (3 - x)e^x$ for all $x \geq 0$
 (C) $\frac{x^2}{2} + 2x + 3 \geq (3 - x)e^x$ for all $x \geq 0$ (D) $\frac{x^2}{2} + 2x + 3 \leq (3 - x)e^x$ for all $x \geq 0$

MMN140

Multiple Correct Question

6. If $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ where $0 < x < 1$, then

- (A) ' f ' is increasing function (B) ' g ' is decreasing function
 (C) $\frac{f(x)}{g(x)}$ is increasing function (D) $g(f(x))$ is decreasing function

MMN141

ANSWER KEY

Question	1	2	3	4	5	6	
Answer	C	D	B	A	C	A,B,C	



Monotonicity : RACE-03

Single Correct Question

1. Exhaustive set of values of 'a' for which the function $f(x) = x^4 + \frac{3x^2}{2} + ax^3 + 1$ will be concave upward along the entire real line, is :
 (A) $[-1, 1]$ (B) $[-2, 2]$ (C) $[0, 2]$ (D) $[0, 4]$
MMN142
2. For which interval, the function $\frac{x^2-3x}{x-1}$ satisfies all the conditions of Rolle's theorem
 (A) $[0, 3]$ (B) $[-3, 0]$ (C) $[1.5, 3]$ (D) For no interval
MMN143
3. Rolle's theorem is not applicable to the function $f(x) = |x|$ defined on $[-1, 1]$ because
 (A) f is not continuous on $[-1, 1]$ (B) f is not differentiable on $(-1, 1)$
 (C) $f(-1) \neq f(1)$ (D) $f(-1) = f(1) \neq 0$
MMN144
4. The function $f(x) = x(x+3)e^{-(1/2)x}$ satisfies all the conditions of Rolle's theorem in $[-3, 0]$. The value of c is
 (A) 0 (B) -1 (C) -2 (D) -3
MMN145
5. Rolle's theorem is true for the function $f(x) = x^2 - 4$ in the interval
 (A) $[-2, 0]$ (B) $[-2, 2]$ (C) $\left[0, \frac{1}{2}\right]$ (D) $[0, 2]$
MMN146
6. For the function $f(x) = e^x$, $a = 0$, $b = 1$, the value of c in Lagrange's mean value theorem will be
 (A) $\log x$ (B) $\log(e-1)$ (C) 0 (D) 1
MMN147
7. If $f(x) = \cos x$, $0 \leq x \leq \frac{\pi}{2}$, then the real number 'c' of the Lagrange's mean value theorem is
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\sin^{-1}\left(\frac{2}{\pi}\right)$ (D) $\cos^{-1}\left(\frac{2}{\pi}\right)$
MMN148
8. From Lagrange's mean value theorem $f(b) - f(a) = (b-a)f'(x_1)$; $a < x_1 < b$ if $f(x) = \frac{1}{x}$, then $x_1 =$
 (A) $\sqrt{ab}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\frac{b-a}{b+a}$
MMN149
9. For the function $x + \frac{1}{x}$, $x \in [1, 3]$, the value of c for the Lagrange's mean value theorem is
 (A) 1 (B) $\sqrt{3}$ (C) 2 (D) None of these
MMN150

10. Let $f(x)$ satisfy all the conditions of Lagrange's mean value theorem in $[0, 2]$.
If $f(0) = 0$ and $|f'(x)| \leq \frac{1}{2}$ for all x , in $[0, 2]$ then
(A) $f(x) \leq 2$ (B) $|f(x)| \leq 1$
(C) $f(x) = 2x$ (D) $f(x) = 3$ for at least one x in $[0, 2]$ MMN151
11. In the Lagrange's mean value theorem $\frac{f(b)-f(a)}{b-a} = f'(c)$, if $a = 0, b = \frac{1}{2}$ and $f(x) = x(x-1)(x-2)$, the value of c is
(A) $1 - \frac{\sqrt{15}}{6}$ (B) $1 + \sqrt{15}$ (C) $1 - \frac{\sqrt{21}}{6}$ (D) $1 + \sqrt{21}$ MMN152
12. The abscissa of the points of the curve $y = x^3$ in the interval $[-2, 2]$, where the slope of the tangents can be obtained by Lagrange's mean value theorem for the interval $[-2, 2]$, are
(A) $\pm \frac{2}{\sqrt{3}}$ (B) $\pm \sqrt{3}$ (C) $\pm \frac{\sqrt{3}}{2}$ (D) 0 MMN153
13. The function $f(x) = (x-3)^2$ satisfies all the conditions of Lagrange's mean value theorem in $[3, 4]$. A point on $y = (x-3)^2$, where the tangent is parallel to the chord joining $(3, 0)$ and $(4, 1)$ is
(A) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (C) $(1, 4)$ (D) $(4, 1)$ MMN154
14. In the Lagrange's mean value theorem, $f(b) - f(a) = (b-a)f'(c)$ if $a = 4, b = 9$ and $f(x) = \sqrt{x}$ then the value of c is
(A) 8.00 (B) 5.25 (C) 4.00 (D) 6.25 MMN155
15. Let Rolle's theorem holds for $f(x) = x^3 + bx^2 + ax$, when $1 \leq x \leq 2$ at the point $c = \frac{4}{3}$, then find $a + b$.
(A) 1 (B) 2 (C) 3 (D) 4 MMN156

ANSWER KEY

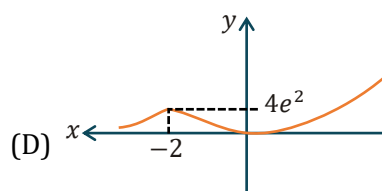
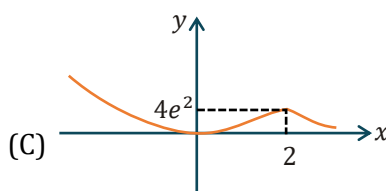
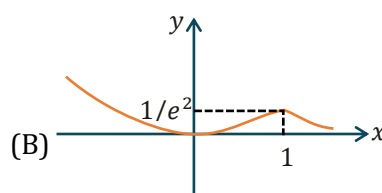
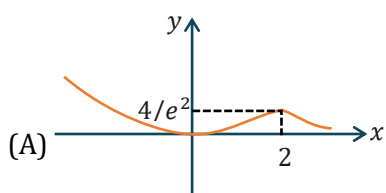
Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	B	C	B	B	C	A	B	B
Question	11	12	13	14	15					
Answer	C	A	B	D	C					



Monotonicity : RACE-04

Single Correct Question

1. Draw graph of $y = x^2 e^{-x}$.



MMN157

Subjective Type Question

2. Draw graph of $f(x) = x \ln x$

MMN158

3. Draw graph of $f(x) = \frac{x}{\ln x}$

MMN159

4. Draw graph of $f(x) = \frac{2x-3}{x+1}$

MMN160

5. Draw graph of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$

MMN161

ANSWER KEY

Question	1
Answer	A